Desarrollo de una herramienta en Python para el postprocesado de modelos térmicos realizados con ESATAN-TMS

Trabajo de Fin de Grado
ESPECIALIDAD: Propulsión Aeroespacial

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Contents

1. Abstract **1**

2. Introduction **3**

2.1. Objectives 3

3. State of Art **5**

3.1. Thermal Tools in the Aerospace Industry 5

3.2. Post-Processing Capabilities in ESATAN-TMS 5

3.3. Custom Interfaces: JPL Example 5

3.4. Practical Application: FalconSat-2 5

3.5. Motivation for the Present Work 5

4. Theoretical Background **7**

4.1. Thermal Control in Engineering 7

4.2. Space Thermal Environment 8

4.3. Heat Transfer Mechanisms 8

4.3.1. Conduction Heat Transfer 8

4.3.2. Radiation Heat Transfer 8

4.4. Mathematical Modeling of Thermal Systems 8

4.4.1. Disadvantages of the Finite Element Method (FEM) 8

4.4.2. Lumped Parameter Method (LPM) 8

4.5. Software and Advanced Methods for Thermal Simulation 10

4.6. Analytical Capabilities of ESATAN-TMS 10

5. Development of the Thermal Model in ESATAN-TMS and Data Export/Analysis 11
5.1. Subsystem Classification . . . 11
5.2. Materials and Properties . . . 11
5.2.1. Thermo-optical properties . . . 11
5.3. Steady-state radiative case . . . 11
5.4. Boundary Conditions and Loads Definition . . . 11
5.5. Model Verification: Group-Level Energy Balances . . . 11
5.6. Data provided by the TMD . . . 11
5.6.1. Use in the Python post-processing . . . 11

6. Design and Implementation of the Post-Processing Code 13
6.1. Code Structure and Architecture . . . 13
6.1.1. Motivation for a Modular Architecture . . . 13
6.1.2. Files overview . . . 13
6.2. Software Stack and Dependencies . . . 13
6.3. Filtering and Grouping Methods . . . 13
6.4. Figure Conventions and Representation Criteria . . . 13
6.5. Scope of Examples and Cryogenic-Flow Validation . . . 13

7. Results and Visualisations from the Post-Processing Workflow 15
7.1. Filtering Results . . . 15
7.2. Sankey Diagram . . . 15
7.3. Network Graph Views (NX) . . . 15
7.4. Stacked-Bar Balances . . . 15
7.5. Cryogenic-Flow Views (ARIEL) . . . 15

8. Conclusions and Future Work 17
8.1. Conclusions . . . 17
8.2. Future Work . . . 17

References 19

Anexos 21

List of Figures

4.1. Simplified heat transfer system and its corresponding thermal network [1] 9

List of Tables

Acronyms

Capítulo 1

Abstract

This Bachelor's Thesis focuses on implementing post-processing tools for the thermal analysis of space systems, using results obtained from the ESATAN-TMS software. The main objective is to obtain accurate data and representation of heat flows performing a transient analysis, based on a previous similar study in a steady-state case.

Capítulo 2

Introduction

2.1. Objectives

Capítulo 3

State of Art

3.1. Thermal Tools in the Aerospace Industry

3.2. Post-Processing Capabilities in ESATAN-TMS

3.3. Custom Interfaces: JPL Example

3.4. Practical Application: FalconSat-2

3.5. Motivation for the Present Work

Capítulo 4

Theoretical Background

4.1. Thermal Control in Engineering

In engineering applications, thermal control is decisive to optimize the performance of the system and its components, while keeping them within temperature limits. In some systems and elements, such as engines, electric components or aeroespace systems, the implementation of a Thermal Control System (TCS) is critical because thermal variations can produce a negative impact on their performance and structural safety [2].

Methods such as convection, conduction and radiation can be implemented on Earth's surface to study heat transfer phenomena. However, the lack of an atmosphere in space, limits heat transfer mechanisms to conduction in solids and thermal radiation into space [3].

The primary objective of a Spacecraft TCS is to ensure that all onboard components operate within their specified temperature ranges during all mission phases, while minimizing resource consumption (mass and power) [4]. This is achieved through a careful balance of internal heat dissipation and external environmental fluxes, utilizing both passive and active thermal control technologies [2].

Those aspects are critical because, unlike terrestrial applications, the space environment is characterized by extreme conditions, such as vacuum, a background temperature of 3K, and rapid, intense thermal cycling due to exposure to sunlight and Earth's shadow [3, 4].

Failure to comply with these thermal limits can lead to equipment malfunctions, accelerated material degradation, reduced reliability, structural damage from thermal cycling, or even catastrophic failure.

The objective of thermal control is to prevent those situations, ensuring operational reliability and enhancing the efficiency of the system [5]. Some key requirements for the TCS include [6]:

- Temperature Ranges: maintaining specific operational and survival temperature ranges for components: electronics, batteries or infrared detectors

- Thermal Gradients and Stability: controlling temperature gradients and stability is essential for high-precision systems such as cameras, optical benches, and CCD sensors to avoid structural distortions and measurement errors

4.2. Space Thermal Environment

Spacecraft thermal control is a process of energy management in which environmental heating plays a major role and therefore, has a major impact in the spacecraft design. There is no convection in space due to the absence of an atmosphere, being radiation the only possible method to dissipate heat in space.
Nevertheless, heat conduction between the structure and the components of the spacecraft still occurs. [2]

4.3. Heat Transfer Mechanisms

Heat can be defined as the form of energy that can be transferred from one system to another as a result of temperature difference. The science that deals with the determination of the rates of such energy transfers is heat transfer [5]. Conduction, convection and radiation are the fundamental mechanisms of heat transfer. However, as it is mentioned earlier, there is no convection in space systems, leaving conduction and radiation as the predominant mechanisms. [2]

4.3.1. Conduction Heat Transfer

d

4.3.2. Radiation Heat Transfer

4.4. Mathematical Modeling of Thermal Systems

Mathematical modeling is essential to predict the thermal behavior of space systems and design effective thermal control strategies.

4.4.1. Disadvantages of the Finite Element Method (FEM)

4.4.2. Lumped Parameter Method (LPM)

The Lumped Parameter Method (LPM) is a discretization technique that simplifies a continuous physical system into a finite number of discrete elements called nodes. The fundamental assumption

is that each node is isothermal, meaning its temperature is uniform throughout its entire volume. This allows the system to be solved as a network of interconnected thermal resistances and capacitances, analogous to an electrical circuit, where temperature equates to voltage and heat flow to current flow. [7]

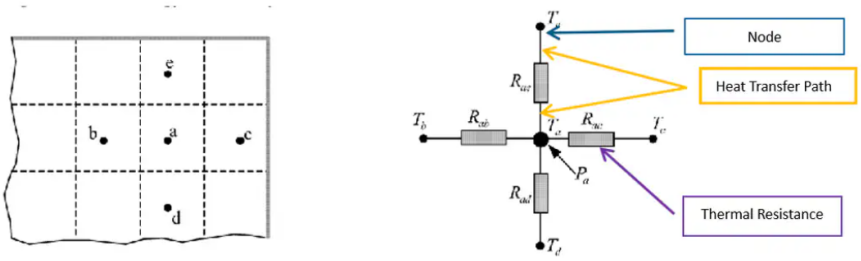

Figura 4.1: Simplified heat transfer system and its corresponding thermal network [1]

The behaviour of the network can be described as a system of differential equations whose coefficients are grouped values of the nodes. The mathematical expression for the nodal energy balance equation is:

$$C_i \frac{dT_i}{dt} = Q_i - \sum_{j=1}^{n} GL_{ij}(T_i - T_j) - \sum_{j=1}^{n} GR_{ij}(T_i^4 - T_j^4) \tag{4.1}$$

where:

- C_i [J/K]: thermal capacitance of node i ($m_i\ c_{p,i}$); stores/releases heat as T_i changes.
- T_i [K]: temperature of node i.
- t [s]: time.
- GL_{ij} [W/K]: conductive conductance between nodes i and j. The value is zero if there is not a conductive path.
- GR_{ij} [W/K^4]: radiative coupling factor between i and j in the non-linear T^4 form; it embeds σ, view factors and emissivities for diffuse–gray surfaces.
- Q_i [W]: net heat source applied to node i (internal dissipation, heaters, and absorbed external fluxes as modelled in the network).
- $\sum_{j=1}^{n}$: sum over all nodes j, which are thermally connected to i. The terms $GL_{ij}(T_i - T_j)$ and $GR_{ij}(T_i^4 - T_j^4)$ contribute as inflow when the expression in parentheses is negative.

The discrete approximation has advantages in numerical as well as experimental work, and lends itself to simple hand calculations as well as to computerised solution. A continuous system obeys

a differential equation, whereas a networked model obeys a system of algebraic equations, when studying a steady-state case, in a finite number of variables which can be solved by standard techniques. [7] Therefore, this method has a wide range of applications in the aeroespace industry, such as the thermal modeling of satellites, where a balance between accuracy and efficiency is required .

4.5. Software and Advanced Methods for Thermal Simulation

4.6. Analytical Capabilities of ESATAN-TMS

Capítulo 5

Development of the Thermal Model in ESATAN-TMS and Data Export/Analysis

5.1. Subsystem Classification

5.2. Materials and Properties

5.2.1. Thermo-optical properties

5.3. Steady-state radiative case

5.4. Boundary Conditions and Loads Definition

5.5. Model Verification: Group-Level Energy Balances

5.6. Data provided by the TMD

5.6.1. Use in the Python post-processing

Capítulo 6

Design and Implementation of the Post-Processing Code

6.1. Code Structure and Architecture

6.1.1. Motivation for a Modular Architecture

6.1.2. Files overview

6.2. Software Stack and Dependencies

6.3. Filtering and Grouping Methods

6.4. Figure Conventions and Representation Criteria

6.5. Scope of Examples and Cryogenic-Flow Validation

Capítulo 7

Results and Visualisations from the Post-Processing Workflow

7.1. Filtering Results

7.2. Sankey Diagram

7.3. Network Graph Views (NX)

7.4. Stacked-Bar Balances

7.5. Cryogenic-Flow Views (ARIEL)

Capítulo 8

Conclusions and Future Work

8.1. Conclusions

8.2. Future Work

References

[1] Understanding the lumped-parameter thermal network method [online].

[2] D. Gilmore (Ed.), Spacecraft Thermal Control Handbook, Volume I: Fundamental Technologies, 2nd Edition, The Aeroespace Press, El Segundo, CA, 2002.

[3] J. Meseguer, I. Pérez-Grande, A.Sanz-Andrés, Spacecraft Thermal Control, 1st Edition, Woodhead Publishing Limited, Cambridge, UK, 2012.

[4] I. G. de Control Térmico Espacial, Descripción y objetivos del TCS: Control térmico espacial, diapositivas de la asignatura Transferencia de calor y control térmico, material docente de la Universidad Politécnica de Madrid[consultado el 6 de marzo de 2026] (2026).

[5] Y. A. Çengel, A. J. Ghajar, Heat and Mass Transfer: Fundamentals and Applications, 6th Edition, McGraw-Hill, New York, NY, 2020.

[6] I. G. de Control Térmico Espacial, Conducción *i*: Thermal conductivity and heat equation unidimensional, diapositivas de la asignatura Transferencia de calor y control térmico, material docente de la Universidad Politécnica de Madrid[consultado el 6 de marzo de 2026] (2026).

[7] I. E. U. Ltd, ESATAN-TMS Thermal Engineering Manual.

Título del anexo

Aquí puedes meter la información que no sea imprescindible en el cuerpo del trabajo pero si que interese que esté en el documento.